

List of Functions for Exam 1

- `+`, `-`, `*`, `/`, `^` – arithmetic operators
- `sqrt()` – square root
- `log()` – logarithm
- `seq()` – sequence creation
- `rep()` – value/vector repetition
- `sort()` – vector sort
- `length()` – determine vector length
- `[ ]` – vector subset/extraction, list slicing, multiple members
- `sum()` – sum all vector elements
- `prod()` – multiply all vector elements
- `mean()` – averages all vector elements
- `sd()` – takes standard deviation of all vector elements
- `matrix()` – create a matrix
- `rbind()` – create a matrix (bind rows)
- `cbind()` – create a matrix (bind columns)
- `dim()` – get matrix dimensions
- `nrow()` – get number of rows
- `ncol()` – get number of columns
- `[ , ]` – matrix/array/data frame subsetting
- `diag()` – diagonal elements/identity matrix
- `array()` – create an array
- `list()` – create a list
- `[[ ]]` – unnamed member reference list
- `$` – get named member/variable
- `data.frame()` – create data frames
- `==`, `!=`, `>`, `<`, `>=`, `<=` – relational operators
- `any()` – checks whether any entries are true
- `all()` – checks whether all entries are true
- `&&`, `&`, `||`, `|`, `!` – logical operators
- `which()` – determines indices of TRUEs
- `nchar()` – gets the number of characters in a string
- `cat()` – concatenates strings (no return)
- `paste()` – pastes strings (returns a string)
- `\` – escape character
- `substr()` – Subsets a string
- `sub()`, `gsub()` – string matching and replacement
- `factor()` – creates a factor vector
- `levels()` – gets levels of a factor
- `cut()` – creates factor from continuum
- `is.infinite()` – element-wise check for Inf
- `is.finite()` – element-wise check for finiteness
- `NaN` – value for invalid numerics
- `is.nan()` – element-wise check for NaN
- `NA` – value for missing observation
- `is.na()` – element-wise check for NA or NaN
- `na.omit()` – delete all NAs and NaNs
- `NULL` – value for empty
- `is.null()` – check for NULL
- `attributes()` – list explicit attributes
- `attr()` – obtain specific attribute
- `dimnames()` – get array dimension names
- `class()` – get object class
- `is._()` – object-checking functions
- `as._()` – object-coercion functions
- `unlist()` – object-coercion from list to vector
- `runif()` – draws pseudo-random numbers from a uniform distribution
- `sample()` – draws numbers from a sample given by user